



Task 1

Digital Engineering Design Group Project

Computational and Software Techniques in Engineering

MSc Degree

Academic Year: 2025 – 2026

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1. Analysis of the strengths/development opportunities of my personal profile

According to Appendix A, the most important weakness is the lack of communication (examples: hesitant to seek support, reluctance to delegate tasks). The best way to transform that into an opportunity is to create a moment during meetings to share difficulties and to split tasks efficiently. The two other themes which can also be improved are the social aspect and the adaptability. They contain more weaknesses than strengths, but the latter have more impact (examples: authentic sincerity, respecting autonomy, openness to challenging ideas). This project will provide me opportunities to develop more facilities in project management.

Appendix A also describes strengths in independence, logic, and structure themes (examples: self-directed learning, innovative mindset, conceptual thinking, objective judgement, continuous improvement, and self-discipline). These qualities can help the group to maintain a good level of energy. On a personal level, these strengths contribute to making me confident to finish tasks and help others.

Appendix B points out that I don't have skills in poster design. The skills hub from the library will help me to learn it. With the others' primary and secondary strengths, I can help with organisation and completing various tasks. Moreover, Appendix C confirms these affirmations.

My main strengths are being able to think logically and plan things out in a structured way. These will help me plan my projects and designs. They could improve in a few areas, such as their capacity to adapt and get along with others. They may discover it more difficult to collaborate and be receptive to new ideas as a result. Although I think these traits will help me with organisational and analytical tasks, I still need to improve the way I communicate with my teammates. I want to perform better if I'm in charge of task assignment. In order to fill in the gaps, I also want to learn how to create posters using simple design tools. Based on this analysis, I want to take on jobs that require me to work with others to get creative work done.

2. Initial proposal of role(s) for the group project

Considering the analysis of my personal profile and the appendices, the following roles can be a great opportunity to the group to benefit from my strengths and to me to develop my skills. They are listed from the most liked to the less.

1. Finisher
2. Organiser
3. Investigator
4. Evaluator

3. Appendices

Appendix A: Personality test

The test was carried out on 16personalities website: [16personalities_report_hinot](#).

The personality found is:

- Profile: Architect (INTJ-T)
- Role: Analysts
- Strategy: Constant Improvement

The principal traits are:

- 82% introverted
- 52% intuitive
- 64% thinking
- 51% judging
- 79% turbulent

All the strengths and weaknesses can be assigned to the following themes: independence, logic, structure, social, and adaptability.

The percentage of each trait, according to the strengths and weaknesses in each theme, is taken and added to obtain the personality weight.

Table 1: Analysis of the personality test's results

Theme	Strengths		Weaknesses		Personality weight	Strength score weighted	Weakness score weighted
	Traits	Score	Traits	Score			
Independence	Independent worker, Self-directed learning	2	Hesitant to seek support, Reluctance to delegate tasks	2	1.61 (Introversion + Turbulent)	3.22 50%	3.22 50%
Logic	Innovative mindset, Conceptual thinking, Objective judgment,	5	Ignoring social dynamics, Critical communication	2	1.16 (Intuitive + Thinking)	5.8 71.43%	2.32 28.57%

	Reflective insight, Insightful advice						
Structure	Continuous improvement, Self-disciplined approach, Clarity of purpose, Reliable performance, Selective loyalty	5	Too rigid expectations, Reluctant to celebrate progress, Neglecting leisure or rest	3	1.3 (Judging + Turbulent)	6.5 62.5%	3.9 37.5%
Social	Authentic sincerity, Meaningful conversation, Quietly caring, Respecting autonomy	4	Insensitive to emotions, Discomfort with networking, Avoiding social rituals, Withdrawn under stress, Difficulty sharing vulnerabilities, Overly blunt feedback, High relationship standards	7	1.46 (Introverting + Thinking)	5.84 36.36%	10.22 63.64%
Adaptability	Openness to challenging ideas, Resilient determination	2	Frustration with constraints, Discomfort with ambiguity, Impatience with routine, Avoiding emotional exploration	4	1.31 (Intuitive + Turbulent)	2.62 33.33%	5.24 66.67%

According to Table 1, independence can be a strength and a weakness at the same time. A special attention will be paid to this matter. The other strengths are structure and logic. There are also good traits in social and adaptability themes but there aren't principal strengths.

Appendix B: Skills Matrix

This skills matrix lists all the competences needed for the project and the level of mastery of them.

Table 2: Skills matrix

Level	Primary Strength	Secondary Strength	Weakness	I have not practised this skill yet
Responding to email	x			
Suggesting ideas to the team		x		
Completing Tasks Efficiently	x			
Helping the team to function (take notes, schedule meetings, book rooms, organise task lists, keep everyone aware of deadlines).	x			
Planning & Organisation	x			
Design of System		x		
Completing Documentation & Reports		x		
Organising meetings	x			
Leadership & Direction		x		
Presenting ideas to customer, team, module leaders		x		
Programming	x			
Graphical Design – GUI Design		x		
Poster Design				x
Testing	x			
Commenting Code	x			
Conflict Resolution		x		
Professional Behaviour	x			

Appendix C: Strengths Ticklist

Taken from *Learning in Teams: A Student Manual*, Graham Gibbs, Oxford Centre for Staff Development, 1994, ISBN 1 873576 20 X, which is in turn based on *Management Teams*, RM Belbin, Heinemann, 1981. In each case put a tick in either the Primary Strength or the Secondary Strength box, as appropriate, or leave blank if you consider yourself not to have any sort of strength in that particular area.

Table 3: Strengths Ticklist

Team Roles	Primary Strength	Secondary Strength
Innovator Produces ideas, imaginative, unorthodox, radical, clever, uninhibited. (Can be over-sensitive, prickly. May need careful handling.)		x
Investigator Finds things out, always knows someone who ..., brings information back to the team, enthusiastic, gregarious. (Can be lazy and complacent.)		x
Chair Self-confident, commands respect, good speaker, thinks positively, good at guiding the team. (Can be domineering, bossy.)		x
Shaper Energetic, drives everyone along, needs to succeed, make things happen. (Can be disruptive and argumentative, impatient and a problem if things don't go their own way.)		x
Evaluator Careful, makes intelligent judgements, tests out ideas, evaluates proposals, helps the team avoid mistakes. (Can become isolated and aloof, pessimistic or over-critical.)	x	
Teamworker Sympathetic, understanding, sensitive, shows a strong concern for social interaction, leads from behind. Places the team above personal concerns. (May be indecisive.)		x
Organiser Methodical, hard-working, reliable, orthodox, turns ideas into plans which are feasible and gets down to tasks which need doing. (Can be inflexible and uninspiring.)	x	
Finisher Painstaking, conscientious, follows through and works hard to finish things properly. Meets deadlines and pays attention to detail. (Can be over-anxious and perfectionist.)	x	

MSc in Computational & Software Techniques in Engineering (DED)

Martin Harambat

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1 Strength analysis

The objective of this first assignment is to identify which role(s) would best suit me for the group project. To achieve this, I completed an online questionnaire on 16Personalities, for which I produced a summary of the test results (Appendix A), and filled out a Skills Matrix (Appendix B) and a Strengths Checklist (Appendix C). With these resources, the aim is to identify my strengths, weaknesses, and development opportunities.

On my personality test (Appendix A), I'm an ENTP-A, the "Innovator." This means I am an Extraverted (57%), Intuitive (77%), Thinking (64%), Prospecting (72%), and Assertive (67%) person.

This profile describes some essential qualities for a group project. As an Extravert, I am a charismatic and persuasive communicator, capable of rallying others around a vision. My Intuitive and Prospecting sides make me an innovative thinker, constantly generating new ideas and seeing potential where others see problems. I am a quick learner and a natural improviser, approaching challenges with enthusiasm. My Thinking side means I excel at strategic planning and rationalizing thoughts, focusing on facts for strategic decisions.

However, my ENTP-A profile also reveals several weaknesses. My love for theory and brainstorming means I can lose interest when it comes to implementation. The "mundane details" of routine tasks can feel stifling, which can lead to procrastination or overlooking responsibilities. I can be easily distracted by new ideas. My tendency to be overly critical of rules and hierarchies might also cause friction with colleagues or authority figures.

These weaknesses can be transformed into improved opportunities. To counter my lack of organization and follow-through, I will use management tools like Trello or a Gantt diagram to structure the project. I can also delegate repetitive tasks to more methodical members. A key development area is to balance my intellectual honesty with empathy and tact, improving my emotional intelligence so that my debates remain productive and not argumentative.

The Skills Matrix (Appendix B) and the Strengths Checklist (Appendix C) have several primary strengths: suggesting ideas, designing systems, providing leadership, and resolving conflicts. So that demonstrates my capacity to innovate, and my side turned toward others. My secondary strengths include organization and programming. However, two weaknesses stand out: completing documentation and commenting code. That suits my profile ENTP-A. My strength also suits my profile with a leadership direction. I like creative and strategic roles. I rely on my colleagues to help me improve my weak points.

2 My role

Through this introspection to make the best role proposition for my group. I want to be a good member, and my abilities can help as effectively as possible. But I want to improve myself and take the team's advice and use this project as a mechanism to progress. From my preferred role to my worst:

- Investigator
- Innovator
- Evaluator
- Organizer
- Shaper

3 Appendices

3.1 Appendices A

Test carried out on <http://www.16personalities.com>

I do a summary of my personality test

3.1.1 Personality

- **Profile:** Innovateur (ENTP-A)
- **Role:** Analysts
- **Strategy:** People Mastery

3.1.2 Traits

- 57% Extraverted
- 77% Intuitive
- 64% Thinking
- 72% Prospecting
- 67% Assertive

All strengths and weaknesses can be grouped according to the following themes: innovation, adaptability, communication, structure, and social relationships.

The percentage of each trait, according to the strengths and weaknesses in each theme, is considered and added to obtain personality weight.

Table 1 : Personality test summaries

Theme	Strengths Traits	Score	Weaknesses Traits	Score	Personality weight (Calculé)	Strength score weighted	Weakness score weighted
Innovation	"Idea generator, Conceptual thinking, Innovative mindset, Quick learner"	4	"Easily distracted, Reluctant to commit, Impatient with mundane tasks"	3	1.49 (Intuitive 0.77 + Prospecting 0.72)	5.96	4.47 (57.1%) / (42.9%)
Adaptability	"Natural improviser, Flexible, Openness to challenging ideas, Resilient determination"	4	"Frustration with constraints, Discomfort with ambiguity"	2	1.44 (Intuitive 0.77 + Assertive 0.67)	5.76	2.88 (66.7%) / (33.3%)
Communication	"Persuasive communicator, Energizing influence, Insightful advice, Independent thinker"	4	"Critical communication, Insensitive to emotions, Overly blunt, Overly	4	1.21 (Extraverted 0.57 + Thinking 0.64)	4.84	4.84 (50.0%) / (50.0%)

			critical of authority"				
Structure	"Continuous improvement, Clarity of purpose, Reliable performance"	3	"Difficulty staying organized, Dislike of constraints, Too rigid expectations, Neglecting leisure or rest"	4	1.39 (Prospecting 0.72 + Assertive 0.67)	4.17	5.56 (42.9%) / (57.1%)
Social	"Authentic sincerity, Meaningful conversation, Respecting autonomy, Entertaining humor"	4	"Discomfort with vulnerability, Impatient listener, Constant devil's advocate, Escalating conflicts, Struggles with commitment, High relationship standards"	6	1.21 (Extraverted 0.57 + Thinking 0.64)	4.84	7.26 (40.0%) / (60.0%)

According to this table, innovation and adaptability are the main strengths. Communication can be both a strength and a weakness. Special attention should be paid to structure and social relationships, which present more weaknesses than strengths.

3.2 Appendices B

Table 2 : Skills matrix

Level	Primary Strength	Secondary Strength	Weakness	I have not practised this skill yet
Responding to email		x		
Suggesting ideas to the team	x			
Completing Tasks Efficiently	x			
Helping the team to function (take notes, schedule meetings, book rooms, organise task lists, keep everyone aware of deadlines).			x	
Planning & Organisation			x	
Design of System	x			
Completing Documentation & Reports			x	

Organising meetings			x	
Leadership & Direction		x		
Presenting ideas to customer, team, module leaders	x			
Programming		x		
Graphical Design – GUI Design		x		
Poster Design		x		
Testing		x		
Commenting Code		x		
Conflict Resolution		x		
Professional Behaviour		x		

3.3 Appendices C

Taken from *Learning in Teams: A Student Manual*, Graham Gibbs, Oxford Centre for Staff Development, 1994, ISBN 1 873576 20 X, which is in turn based on *Management Teams*, RM Belbin, Heinemann, 1981. In each case put a tick in either the Primary Strength or the Secondary Strength box, as appropriate, or leave blank if you consider yourself not to have any sort of strength in that area.

Table 3 : Ticklist

Team Roles	Primary Strength	Secondary Strength
Innovator Produces ideas, imaginative, unorthodox, radical, clever, uninhibited. (Can be over-sensitive, prickly. May need careful handling.)	x	
Investigator Finds things out, always knows someone who ..., brings information back to the team, enthusiastic, gregarious. (Can be lazy and complacent.)		x

Chair Self-confident, commands respect, good speaker, thinks positively, good at guiding the team. (Can be domineering, bossy.)	X	
Shaper Energetic, drives everyone along, needs to succeed, make things happen. (Can be disruptive and argumentative, impatient and a problem if things don't go their own way.)		X
Evaluator Careful, makes intelligent judgements, tests out ideas, evaluates proposals, helps the team avoid mistakes. (Can become isolated and aloof, pessimistic or over-critical.)		X
Teamworker Sympathetic, understanding, sensitive, shows a strong concern for social interaction, leads from behind. Places the team above personal concerns. (May be indecisive.)	X	
Organiser Methodical, hard-working, reliable, orthodox, turns ideas into plans which are feasible and gets down to tasks which need doing. (Can be inflexible and uninspiring.)		X
Finisher Painstaking, conscientious, follows through and works hard to finish things properly. Meets deadlines and pays attention to detail. (Can be over-anxious and perfectionist.)	X	

MSc in Computational & Software Techniques in Engineering (Digital Engineering Design Option)

Project Group
Task 1



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1 Strength analysis

The objective of this first assignment is to identify which role(s) would best suit me for the group project. To achieve this, I completed an online questionnaire on 16Personalities, for which I produced a summary of the test results (Appendix A), and filled out a Skills Matrix (Appendix B) and a Strengths Checklist (Appendix C). With these resources, the aim is to identify my strengths, weaknesses, and development opportunities.

On my personality test (Appendix A), I'm an ENTP-T. Extravert, Intuitive, Thinking, Prospering and turbulent person. That describe some essential qualities for group project. I'm an extravert. Having contact with people is for me a source of energy. I also express myself openly, so it is good communication skills. My intuitive skill helps me with my imagination and my future opportunities. My thinking side rationalizes my thoughts and focusses on the facts. It is very important for strategic decision. My prospecting profile makes me adaptable and flexible for unforeseen events. So, I can generate ideas, adapt to changes, think long-term and communicate easily with others.

However, my ENTP-T profile also reveals several weaknesses. That requires proactive management. I have some issues for being organized and sometimes which risk overlooking important responsibilities because of that. I am easily distracted by my own ideas. My tendency to be overly critical of rules and the hierarchy so that compromise my relationships with my colleagues or my hierarchy. However, these weaknesses can be transformed into improvement opportunities. Against my lack of organization, I will use management tools like Trello or Gantt diagram for structure and have a view on the project. I can delegate repetitive tasks to more methodical members. I will organize some short, fixed documentation sessions to focus on myself in one direction. I also work on my emotional intelligence problem to balance my honesty with empathy.

The Skills Matrix (Appendix B) and the Strengths Checklist (Appendix C) several primary strengths: suggesting ideas, designing systems, providing leadership, and resolving conflicts. So that demonstrates my capacity to innovate, and my side turned toward others. My secondary strengths include organization and programming. However, two weaknesses stand out: completing documentation and commenting code. That suits my profile ENTP-T. My strength also suits my profile with a leadership direction. I like creative and strategic roles. I rely on my colleague to help me improve my weak points.

2 My role

Through this introspection to make the best role proposition for my group. I want to be a good member, and my abilities can help as effectively as possible. But I want to improve myself and take the team's advice and use this project as a mechanism to progress.

From my preferred role to my worst:

- Investigator
- Innovator
- Shaper
- Evaluator
- Organizer

3 Appendices

3.1 Appendices A

Test carried out on <http://www.16personalities.com>

I do a summary of my personality test

3.1.1 Personality

- **Profile:** Debater (ENTP-T)
- **Role:** Analysts
- **Strategy:** Constant Improvement

3.1.2 Traits

- 78% extraverted
- 64% intuitive
- 85% thinking
- 54% prospecting
- 63% turbulent

All strengths and weaknesses can be grouped according to the following themes: innovation, adaptability, communication, structure, and social relationships.

The percentage of each trait, according to the strengths and weaknesses in each theme, is considered and added to obtain personality weight.

Table 1 : Personality test summaries

Theme	Strengths Traits	Score	Weaknesses Traits	Score	Personality weight	Strength score weighted	Weakness score weighted
Innovation	Idea generator, Conceptual thinking, Innovative mindset, Quick learner	4	Easily distracted, Reluctant to commit, Impatient with mundane tasks	3	1.275 (Intuitive + Prospecting)	5.1 59.30%	3.825 40.70%
Adaptability	Natural improviser, Flexible, Openness to challenging	4	Frustration with constraints, Discomfort with ambiguity	2	1.27 (Intuitive + Turbulent)	5.08 72.57%	2.54 27.43%

	ideas, Resilient determination						
Communication	Persuasive communicator, Energizing influence, Insightful advice, Independent thinker	4	Critical communication, Insensitive to emotions, Overly blunt, Overly critical of authority	4	1.55 (Extraverted + Thinking)	6.2 50%	6.2 50%
Structure	Continuous improvement, Clarity of purpose, Reliable performance	3	Difficulty staying organized, Dislike of constraints, Too rigid expectations, Neglecting leisure or rest	4	1.17 (Prospecting + Turbulent)	3.51 42.86%	4.68 57.14%
Social	Authentic sincerity, Meaningful conversation, Respecting autonomy, Entertaining humor	4	Discomfort with vulnerability, Impatient listener, Constant devil's advocate, Escalating conflicts, Struggles with commitment, High relationship standards	6	1.43 (Extraverted + Thinking)	5.72 40%	8.58 60%

According to this table, innovation and adaptability are the main strengths. Communication can be both a strength and a weakness. Special attention should be paid to structure and social relationships, which present more weaknesses than strengths.

3.2 Appendices B

Table 2 : Skills matrix

Level	Primary Strength	Secondary Strength	Weakness	I have not practised this skill yet
Responding to email		x		
Suggesting ideas to the team	x			

Completing Tasks Efficiently		X		
Helping the team to function (take notes, schedule meetings, book rooms, organise task lists, keep everyone aware of deadlines).		X		
Planning & Organisation		X		
Design of System	X			
Completing Documentation & Reports			X	
Organising meetings	X			
Leadership & Direction	X			
Presenting ideas to customer, team, module leaders	X			
Programming		X		
Graphical Design – GUI Design		X		
Poster Design		X		
Testing		X		
Commenting Code				X
Conflict Resolution	X			
Professional Behaviour		X		

3.3 Appendices C

Taken from *Learning in Teams: A Student Manual*, Graham Gibbs, Oxford Centre for Staff Development, 1994, ISBN 1 873576 20 X, which is in turn based on *Management Teams*, RM Belbin, Heinemann, 1981. In each case put a tick in either the Primary Strength or the Secondary Strength box, as appropriate, or leave blank if you consider yourself not to have any sort of strength in that area.

Table 3 : Ticklist

Team Roles	Primary Strength	Secondary Strength
Innovator Produces ideas, imaginative, unorthodox, radical, clever, uninhibited. (Can be over-sensitive, prickly. May need careful handling.)	X	
Investigator Finds things out, always knows someone who ..., brings information back to the team, enthusiastic, gregarious. (Can be lazy and complacent.)		X
Chair Self-confident, commands respect, good speaker, thinks positively, good at guiding the team. (Can be domineering, bossy.)	X	
Shaper Energetic, drives everyone along, needs to succeed, make things happen. (Can be disruptive and argumentative, impatient and a problem if things don't go their own way.)		X
Evaluator Careful, makes intelligent judgements, tests out ideas, evaluates proposals, helps the team avoid mistakes. (Can become isolated and aloof, pessimistic or over-critical.)		X
Teamworker Sympathetic, understanding, sensitive, shows a strong concern for social interaction, leads from behind. Places the team above personal concerns. (May be indecisive.)	X	

Organiser Methodical, hard-working, reliable, orthodox, turns ideas into plans which are feasible and gets down to tasks which need doing. (Can be inflexible and uninspiring.)		X
Finisher Painstaking, conscientious, follows through and works hard to finish things properly. Meets deadlines and pays attention to detail. (Can be over-anxious and perfectionist.)	X	

Task 1: Individual Report

Digital Engineering Design Group Project
Computational and Software Techniques in Engineering



John Hoarau

Prepared for November 14, 2025

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Abstract

This report presents a personal analysis of my strengths and development opportunities for the 2025-2026 Digital Engineering Design Group Project, as required by Task 1 of the module brief [1]. The analysis is a synthesis of my professional experience from my CV [2], the results of a free test (which identified my profile as ENTJ "Commander") [3], a self-assessed Skills Matrix [4], and the Belbin-based Strengths Ticklist .

Key strengths are identified in leadership, strategic planning, and task execution. Key development opportunities are noted in interpersonal conflict resolution and specific creative skills like poster design. The report concludes with an initial proposal of team roles, such as Chair and Shaper, to best leverage my strengths for the group's success.

1 Analysis of Strengths/Development Opportunities

This analysis is based on my professional CV [2], the ENTJ "Commander" personality profile [3], and the self-assessments in Appendix B and C.

1.1 Strengths

My profile indicates a "natural-born leader" [3] with strong will and a drive for achievement. This is reflected in my CV through projects I have led [2], and confirmed by the Skills Matrix (Appendix B), which shows primary strengths in:

- Leadership & Direction [4]
- Planning & Organisation [4]
- Completing Tasks Efficiently [4]
- Organising meetings [4]
- Presenting ideas [4]

As an ENTJ, I am rational, enjoy a challenge [3], and possess strong analytical skills (e.g., Finite Element Analysis, data analysis) [2], which are well-suited to the project's technical demands.

1.2 Development Opportunities

The most significant development area is interpersonal. The Skills Matrix identifies "Conflict Resolution" as a weakness [4]. This aligns perfectly with the ENTJ profile, which notes a "distance from their emotions" and a tendency to "overlook the emotional sensitivity of individuals," often perceiving emotional displays as weakness [3]. For this team project, I must consciously practice empathy and remember that the team's validation is essential for success [3].

On a technical level, the Skills Matrix also identifies:

- **Poster Design** as a weakness [4].
- **Commenting Code** as a skill I have not yet practised enough [4].

I will need to either learn these skills quickly using university resources or rely on the expertise of other team members in these areas.

2 Initial Proposal of Role(s) for the Group Project

Considering the analysis of my personal profile (see Section 1) and the appendices, the following roles, based on the Belbin framework , are proposed. They are listed from most preferred to least preferred:

1. **Chair:** This role aligns perfectly with the ENTJ "natural-born leader" profile [3], my "Self-confident" nature, and my primary strength in "Leadership & Direction" [4].
2. **Shaper:** This role fits the ENTJ's high drive, determination, and focus on pushing through challenges to achieve goals [3]. My CV demonstrates experience in leading projects to successful completion [2].
3. **Organiser:** This leverages my primary strengths in "Planning & Organisation" and "Organising meetings" [4] to turn the group's ideas into a feasible plan.
4. **Evaluator:** This role would utilise my strong analytical problem-solving skills [2] and the rational, objective judgment of the ENTJ type [3].

A Appendix A: Personality Test Discussion

The test was carried out on 16personalities.com [3], as a free online personality test was required for this task [1].

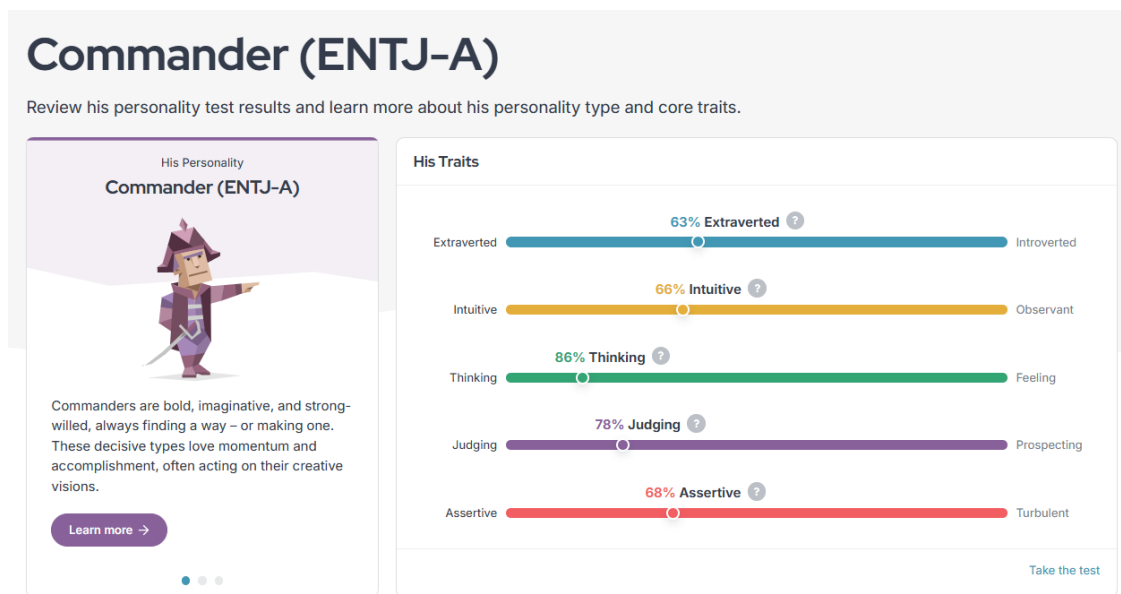


Figure 1: Personality test results for John Hoarau.

The results identified the following profile:

- **Profile:** ENTJ-A (Commander)
- **Role:** Analyst
- **Strategy:** Assertive
- **Trait Percentages:**
 - 63% Extraverted
 - 66% Intuitive

- 86% Thinking
- 78% Judging
- 68% Assertive

Thematic Analysis of Personality

The table below follows the methodology of my teammates to break down the profile traits into weighted themes.

Table 1: Thematic Analysis of ENTJ-A Personality Results

Theme	Strengths Traits	Score	Weaknesses Traits	Score	Personality weight (Traits)	Strength score	Weakness score
Leadership & Drive	"Natural-born leader" [3], Strong-willed [3], Charisma and confidence [3], Drive and determination [3]	4	Dominant and unforgiving [3], Intensity can rub people the wrong way [3]	2	1.41 (Extraverted + Judging)	5.64 71.4%	2.82 28.6%
Strategic Thinking	"Think strategically" [3], "Hold a long-term focus" [3], "Construct creative visions" [3], "Find a way, or make one" [3]	4	Impatient with those who can't keep up [3]	1	1.52 (Intuitive + Thinking)	6.08 80.0%	1.52 20.0%
Rational Objectivity	"Ruthless rationality" [3], "Sharp mind" [3], "Skill in recognizing talents" [3]	3	"Chilling insensitivity" [3], "Overlook emotional sensitivity" [3], "See emotions as weakness" [3]	3	1.54 (Thinking + Assertive)	4.62 50.0%	4.62 50.0%
Interpersonal	"Draws crowds together" [3], "Sensitive to validation/feedback" [3], Assertive	3	"Distance from emotions" [3], "Easy to make enemies" [3], "Conflict Resolution weakness" [4]	3	1.31 (Extraverted + Assertive)	3.93 50.0%	3.93 50.0%

Discussion

The ENTJ "Commander" personality type is defined by natural leadership, confidence, and a powerful drive to achieve goals [3]. My strengths lie in strategic thinking and "ruthless rationality," allowing me to create visions and execute them with determination [3]. This is reflected in my CV through successes in leading teams and engineering complex systems [2].

The primary weakness for this personality type is a "distance from their emotions" [3]. ENTJs can be insensitive, overlook the emotional needs of others, and perceive emotional expression as weakness [3]. This is a significant risk in a team environment. For this group project, I must make a conscious effort to recognize the contributions and needs of my teammates from an emotional standpoint [3].

B Appendix B: Skills Matrix

Table 2: Self-assessed skills matrix [4]

Skill	Primary Strength	Secondary Strength	Weakness	I have not practised this skill yet
Responding to email		x		
Suggesting ideas to the team		x		
Completing Tasks Efficiently	x			
Helping the team to function...	x			
Planning & Organisation	x			
Design of System		x		
Completing Documentation & Reports		x		
Organising meetings	x			
Leadership & Direction	x			
Presenting ideas...	x			
Programming	x			
Graphical Design – GUI Design		x		
Poster Design			x	
Testing		x		
Commenting Code				x
Conflict Resolution			x	
Professional Behaviour	x			

C Appendix C: Strengths Ticklist

Table 3: Belbin-based strengths ticklist, adapted from Gibbs.

Team Roles	Primary Strength	Secondary Strength
Innovator Produces ideas, imaginative, unorthodox...		x
Investigator Finds things out, brings information back...		x
Chair Self-confident, commands respect, good speaker...	x	
Shaper Energetic, drives everyone along, needs to succeed...	x	
Evaluator Careful, makes intelligent judgements...	x	
Teamworker Sympathetic, understanding, sensitive...		
Organiser Methodical, hard-working, reliable, turns ideas into plans...	x	
Finisher Painstaking, conscientious, follows through...		x

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Individual Reflection Report

Task 8 (Individual Report)



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1 Introduction

This report reflects on my personal experience as Project Lead of Group Chenonceau throughout the Digital Engineering Design Group Project, undertaken as part of the MSc in Computational and Software Techniques in Engineering at Cranfield University. The project spanned from November 2025 to May 2026 and encompassed nine sequential tasks, progressing from individual personality profiling and team planning through to the implementation and evaluation of advanced optimisation algorithms applied to a real-world structural engineering problem.

The following sections evaluate the accuracy of my Task 1 personality assessment in light of the project experience, detail my personal contributions, reflect on the lessons I will carry into my career, review the effectiveness of our Agile methodology, and provide an honest analysis and peer assessment of each group member. Timesheets are attached as Appendix A.

2 Personality Profile Reflection

2.1 Task 1 Assessment

In Task 1, I was identified as an **ENTJ** — the “Commander” type within the Myers-Briggs Type Indicator (MBTI) framework — and assigned the Belbin Team Roles of *Chair* (now “Co-ordinator”) and *Shaper*. The ENTJ profile characterises individuals as decisive, goal-oriented, and naturally drawn to leadership; individuals who organise people and resources toward clear objectives, maintain momentum under pressure, and are comfortable making difficult decisions. On paper, this categorisation aligned well with the role I was assigned. In hindsight, the fit was largely accurate, though the lived reality of leading a real project revealed layers of complexity that a static personality test cannot fully anticipate.

2.2 How the Profile Held Up in Practice

The core ENTJ traits manifested consistently throughout the project. From the very first meeting, I took ownership of the group’s direction: I chaired every Scrum session, authored the Task 2 planning report, allocated roles based on individual strengths, and served as the primary liaison with the module leader. When decisions needed to be made

— about algorithm selection, report structure, or task prioritisation under deadline pressure — I made them clearly and communicated them to the group. This decisiveness, which the ENTJ profile predicts, proved genuinely valuable in a distributed team environment where ambiguity and inaction are costly.

The Shaper aspect of my Belbin profile also manifested visibly. I consistently pushed the group to maintain quality and pace, and I was not reluctant to raise uncomfortable points when progress was stalling. In the convergence of Tasks 6 and 7, for example, when deliverable timelines were tightest, I made explicit decisions about what to prioritise and held the team to those decisions.

However, the project also surfaced the limitations of the ENTJ archetype. One of the well-documented weaknesses of this type is impatience, combined with a tendency to take on too much personally rather than delegate effectively. This pattern played out clearly over the course of the project. As both a leader and a significant technical contributor — responsible for code integration, LaTeX report writing, and project coordination — I found it difficult at times to separate the two roles. The boundary between what I should direct and what I should do myself became blurred, particularly in the integration phases of Tasks 6 and 7, where I absorbed tasks that fell between individual responsibilities.

Furthermore, the reality of a geographically distributed team — members split between Cranfield (UK) and ESTIA (France) — demanded a degree of interpersonal sensitivity and adaptability that pure strategic decisiveness cannot provide. Maintaining engagement across time zones, different academic schedules, and varying working styles required me to adjust my communication approach continuously in ways that the ENTJ label alone does not describe.

2.3 Summary

The ENTJ profile proved to be a sound starting point: it correctly predicted my leadership instincts, my comfort with decision-making, and my drive for outcomes. What it could not capture was the situational nature of leadership — the moments where a leader must step back, listen more carefully, or redistribute load rather than absorb it. Those are lessons I have drawn directly from this project, and they are more granular than any personality test can convey.

3 Personal Contributions to the Project

My contributions to Group Chenonceau were broad in scope and spanned the technical, organisational, and documentary dimensions of the project throughout all tasks.

3.1 Project Leadership and Coordination

From Task 2 onward, I was responsible for the overall management of the group. This included chairing all weekly Scrum meetings on Microsoft Teams, maintaining and updating the Notion Kanban board, assigning tasks and tracking their completion, managing the GitHub repository structure, and ensuring that no deliverable reached its submission date without the necessary components assembled and reviewed. In a distributed team, this coordination role is substantive and continuous: it involves sending reminders, resolving ambiguities, managing sequencing between contributors, and absorbing last-minute changes without propagating panic to the rest of the team.

3.2 Technical Integration

My most technically demanding contribution was the code integration work for Task 6, the project's culmination. Each group member developed their algorithm independently — Martin implementing BFGS, Pierre implementing Simulated Annealing, and Clémence implementing Particle Swarm Optimisation — but the Task 6 deliverable required all three to operate on the same problem, with consistent interfaces, a shared penalty-based constraint formulation, and a unified benchmarking and results framework. I designed and implemented this integration architecture, which required a thorough understanding of each algorithm's inputs and outputs, the definition of a common objective function interface, and extensive validation to ensure that results were reproducible and comparable across all eight study cases.

3.3 Documentation and Report Writing

Across multiple tasks, I contributed substantially to L^AT_EX report writing — not only authoring sections but also maintaining consistency of style, notation, and structure across collaboratively written documents. The group's reports consistently achieved a professional standard, and sustaining that standard across contributors with varying

writing and formatting experience required dedicated effort at the editing and integration stage.

3.4 Collective Working Culture

It is important to acknowledge that in Group Chenonceau, the boundary between individual and collective work was deliberately permeable. While primary responsibilities were clearly assigned, members routinely supported one another across domains. I actively encouraged this collaborative approach, and it was a key reason why the group's outputs consistently exceeded what any individual could have produced alone.

4 Lessons Learned and Career Development

4.1 Workload Management

The most significant personal lesson from this project concerns workload management. As project leader and a core technical contributor, I repeatedly found myself absorbing tasks that fell between formal responsibilities — integration work that could not be easily assigned to others, report sections requiring a consistent authorial voice, or last-minute revisions before submission. The result was that my own workload peaked sharply around major deadlines, creating unnecessary pressure.

Going forward, I need to be more deliberate in defining the scope of my role at the outset of any project, more explicit in redistributing floating tasks, and more willing to accept “good enough” outputs from collaborators rather than defaulting to doing it myself. The cost of absorbing tasks quietly is not just personal stress — it also prevents team members from developing the ownership and skill they would gain by working through difficulties themselves.

4.2 Communication Precision

The second important lesson concerns communication. While I maintained structured group communication throughout the project, I sometimes assumed shared understanding of my expectations — particularly around document formatting, code style, and integration requirements — when I had not stated those expectations explicitly. In future projects, I will invest more time at the start of each task in articulating the “definition of

done” clearly, so that contributors can work independently with confidence rather than requiring multiple rounds of review.

4.3 Career Direction

This project has meaningfully clarified my career direction. I am currently undertaking an internship at Alten in an R&D context, but my experience as project leader has revealed that I find the greatest satisfaction in roles that combine technical problem-solving with team leadership and tangible, visible delivery. Pure research and development, with its long feedback loops and incremental progress, does not align well with my drive for concrete results and measurable impact — a disposition that the ENTJ profile had already flagged, and that the project confirmed in practice.

I am increasingly drawn to roles in technical project management or engineering leadership — ideally in an international context, as I am keen to build a career outside of France, where both the market conditions and the professional recognition for these roles are more developed. This project has given me concrete, demonstrable evidence — well-structured code, publication-quality reports, a successfully coordinated distributed team — that I can operate effectively at that level. That evidence is something I will present actively as I pursue those opportunities.

5 Review of Agile Methodology

5.1 Implementation

Group Chenonceau adopted an Agile methodology as specified in the module brief, operationalised through the following practices: weekly Scrum meetings via Microsoft Teams; a Kanban board in Notion to track tasks through “To Do,” “In Progress,” and “Done” stages; a Gantt chart for milestone planning; task assignment with attributed ownership; and version control via GitHub for all code assets.

5.2 What Worked Well

The most notable success was the regularity and quality of our Scrum meetings. Every session was scheduled, attended (or communicated about in advance), and productive.

When a member could not attend due to scheduling conflicts arising from the split between ESTIA and Cranfield, they received the relevant notes and action points promptly, ensuring that no one fell out of synchronisation. This consistency was a deliberate priority: the meeting would happen, it would be brief, and everyone would leave knowing exactly what was expected of them before the next session.

The Notion Kanban board proved particularly valuable for maintaining shared situational awareness in a distributed team. At any moment, every member could see what was done, what was in progress, and what was blocked — without needing to send status-request messages. This transparency reduced coordination overhead significantly and made the group's collective progress visible in real time.

The GitHub repository provided a structured, version-controlled environment for code collaboration. The use of branches and documented commits meant that integration events — which might otherwise have caused loss of work — were managed cleanly throughout the project.

5.3 Limitations

Agile's iterative, retrospective dimension was less fully realised. Because each Task constituted a discrete deliverable with a hard deadline, the project structure naturally segmented into phases rather than genuinely iterative cycles. The retrospective element — reflecting formally on what worked and what should change before starting the next sprint — was generally absent. We tended to move from one deadline to the next without pausing to consolidate lessons in a structured way.

In future projects of this kind, I would build explicit, brief retrospective sessions into the inter-task periods. Even thirty minutes of structured reflection after each major submission would have helped the group improve its process incrementally rather than waiting until the final reflection report to identify recurring patterns.

5.4 Overall Assessment

Despite this limitation, Agile was clearly the correct methodology for this project. It imposed enough structure to keep a distributed team coordinated across nine tasks and seven months without becoming bureaucratic, and it produced consistent, on-time deliverables throughout. The lightweight nature of our implementation was appropriate

to a four-person academic project, and I would adopt the same approach — with the retrospective addition — for future collaborative work.

6 Group Member Analysis and Peer Assessment

6.1 Martin Harambat

Martin’s primary contribution was the BFGS quasi-Newton algorithm, developed and validated across Tasks 4 and 6. In the Belbin framework, he fulfilled the *Plant/Innovator* role assigned to him in Task 1 with accuracy: he brought creative energy to technical discussions, consistently proposed novel approaches, and was willing to pursue unconventional solutions when the standard ones proved inadequate. His BFGS implementation was technically sound and produced results that integrated cleanly into the Task 6 framework.

In terms of overall contribution, Martin’s engagement was slightly less consistent than that of Pierre and Clémence. There were periods — particularly during high-pressure deadlines — where his deliverables required more active follow-up and integration support than the others’. That said, he always ultimately delivered, and his creative contributions to group discussions added genuine value to the team’s thinking. He is a capable engineer whose output quality is high when fully engaged.

Peer Assessment Score: 4/5

6.2 Pierre Bédrune

Pierre’s primary responsibility was the Simulated Annealing (SA) algorithm, implemented and applied across Tasks 3 and 6. He fulfilled the *Investigator/Evaluator* Belbin role effectively, approaching problems with rigour and consistency. Pierre’s code was clean, well-structured, and thoroughly documented, which significantly reduced the integration burden during the Task 6 assembly phase. He was proactive in meeting participation, clear in identifying blockers, and collaborative in supporting other workstreams once his own contributions were complete.

Pierre’s reliability and technical precision were a consistent asset to the group. He is the kind of contributor who makes a project leader’s job easier: his deliverables arrive in the expected format, on the expected schedule, with the expected quality.

Peer Assessment Score: 5/5

6.3 Clémence-Philomène Hinot

Clémence fulfilled the *Organiser/Finisher* Belbin role identified in Task 1 with exceptional consistency. Her primary responsibilities were Particle Swarm Optimisation (PSO) and L^AT_EX document production. In both areas, she worked with a precision and attention to detail that elevated the standard of every group deliverable.

Clémence was the group's quality anchor. She regularly identified inconsistencies in notation, formatting, and argumentation that others had overlooked, and she held the group to a standard of completeness that prevented premature closure on documents. Her L^AT_EX formatting skills were the strongest in the group, and the professional visual quality of our reports reflects her contribution directly. Her influence on the group's outputs is difficult to quantify numerically but is present on every page of every deliverable.

Peer Assessment Score: 5/5

6.4 Self-Assessment (John Hoarau)

Throughout the project, I fulfilled my role as Project Lead, technical integrator, primary meeting chair, and significant contributor to documentation and code. As detailed in the preceding sections, I also carried weaknesses: a tendency to absorb too much and to communicate expectations implicitly rather than explicitly. These are genuine areas for improvement, and I acknowledge them honestly.

That said, the outcomes of the project provide the most meaningful evidence of contribution. Group Chenonceau delivered consistent, high-quality work across all nine tasks, on time, from a distributed team of four people. That does not happen without sustained coordination and leadership. I am proud of the role I played in making it happen.

Self-Assessment Score: 5/5

6.5 Peer Assessment Summary

Group Member	Score (0–5)	Belbin Role
John Hoarau	5	Chair / Shaper
Martin Harambat	4	Plant / Innovator
Pierre Bédrune	5	Investigator / Evaluator
Clémence-Philomène Hinot	5	Organiser / Finisher

Table 1: Peer assessment scores for Group Chenonceau (scale 0–5).

7 Conclusion

This project has been, in the truest sense, a learning experience — not primarily in optimisation algorithms, though I developed real competence there, but in the practice of leading a team through sustained, complex work. The ENTJ profile from Task 1 captured my instincts accurately but could not prepare me for the situational judgements that real leadership requires. My contributions were substantial and broad, but so were the personal challenges I navigated. The lessons I take from this — on workload management, on communication clarity, on the kind of work that energises me — are ones I intend to carry actively into the next phase of my career.

Group Chenonceau produced work I am genuinely proud of. The cooling tower report stands as the clearest evidence: a publication-quality technical document integrating three independent algorithms on a real engineering problem, produced by a four-person team working across two countries. That achievement belongs to all of us.

A Timesheets

The following table provides an estimated record of hours spent on each task of the group project. These figures are approximate, as time was not formally logged in a dedicated tool throughout the project, and some tasks involved concurrent work across multiple sessions.

Task	Description	Period	Hours
Task 1	Individual Report (Personality & Skills)	Oct–Nov 2025	6
Task 2	Group Planning Report & Gantt Chart	Nov 2025	8
Task 3	PSO & SA Optimisation (coordination + review)	Nov–Dec 2025	12
Task 4	BFGS Optimisation (coordination + integration)	Dec 2025–Jan 2026	14
Task 5	Interim Presentation (slides + delivery)	Jan 2026	10
Task 6	Cooling Tower Optimisation (code + integration + report)	Jan–Mar 2026	35
Task 7	ABAQUS Task (coordination + report)	Mar 2026	16
Task 8	Individual Reflection Report	Apr–May 2026	8
Task 9	Final Presentation (preparation + delivery)	Apr–May 2026	12
Total			121

Table 2: Timesheet: John Hoarau — estimated hours per task.